

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Classified Practice

Coordinate Geometry Basics

Q001 **Written****A (−4), B (8): distance AB**

Q002 Written**A (−5), B (7): distance AB**

Classified Practice

Coordinate Geometry Basics

Q003 **Written****For two points A (-2) and B (6) , find the distance AB.**

Q004 Written

For two points A (4) and B (10), find the distance AB.

Classified Practice

Coordinate Geometry Basics

Q005 MCQ

Find A (−4), B (8): distance AB:**Choose the correct option.**

A 11

B −12

C 12

D 13

Q006 MCQ**Find a (-5) , B (7) : distance AB:****Choose the correct option.**

A 11

B -12

C 12

D 13

Classified Practice

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Choose the correct option.**

A 7

B -8

C 8

D 9

Q008 MCQ**For two points A (4) and B (10), find the distance AB:****Choose the correct option.**

A 5

B -6

C 6

D 7

Classified Practice

Coordinate Geometry Basics

Q009 **Written****A (−4), B (8): internal point C in ratio 2:1**

Q010 Written**A (-5) , B (7) : internal point C in ratio 2:1**

Classified Practice

Coordinate Geometry Basics

Q011 Written

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Q012 Written

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Classified Practice

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Choose the correct option.**A -4 B 4 C 5 D 3

Q014 MCQ**Find a (-5) , B (7) : internal point C in ratio 2:1:****Choose the correct option.**

A 2

B -3

C 3

D 4

Classified Practice

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Choose the correct option.

A -4

B 4

C 5

D 3

Q016 MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Choose the correct option.

A -8

B 8

C 9

D 7

Classified Practice

Coordinate Geometry Basics

Q017 **Written****A (−4), B (8): external point D in ratio 2:3**

Q018 Written**A (−5), B (7): external point D in ratio 2:5**

Classified Practice

Coordinate Geometry Basics

Q019 **Written**

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Q020 Written

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Choose the correct option.**A -28 B -27 C -29 D 28

Q022 MCQ**Find a (-5) , B (7) : external point D in ratio 2:5:****Choose the correct option.****A** 13**B** -13 **C** -12 **D** -14

Classified Practice

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Choose the correct option.

A 18

B 19

C 17

D -18

Q024 MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Choose the correct option.

A -1

B 1

C 2

D 0

Classified Practice

Coordinate Geometry Basics

Q025 Written

A (−4), B (8): midpoint M

Q026 Written**A (−5), B (7): midpoint M**

Classified Practice

Coordinate Geometry Basics

Q027 **Written**

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.

Q028 Written

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Choose the correct option.**

A 1

B -2

C 2

D 3

Q030 MCQ**Find a (-5) , B (7) : midpoint M:****Choose the correct option.****A** 1**B** 2**C** 0**D** -1

Classified Practice

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:

Choose the correct option.

A 3

B 1

C -2

D 2

Q032 MCQ**For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:****Choose the correct option.****A** 7**B** 8**C** 6**D** -7

Classified Practice

Coordinate Geometry Basics

Q033 **Challenge****Pods A (-8) and B (-3) : fibre-optic tether length**

Classified Practice

Coordinate Geometry Basics

Q034 **Challenge****Pods A (-8) , B (-3) : internal sensor point in ratio 1:4**

Classified Practice

Coordinate Geometry Basics

Q035 **Challenge****Pods A (-8) , B (-3) : external outpost in ratio 3:4**

Classified Practice

Coordinate Geometry Basics

Q036 **Challenge****Pods A (-8) , B (-3) : midpoint supply hub**

Classified Practice

Coordinate Geometry Basics

Q037 **Challenge****Find pods A (-8) and B (-3) : fibre-optic tether length:****Choose the correct option.**

A 5

B 6

C 4

D -5

Classified Practice

Coordinate Geometry Basics

Q038 **Challenge****Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:****Choose the correct option.**A -8 B 7 C -7 D -6

Classified Practice

Coordinate Geometry Basics

Q039 **Challenge****Find pods A (-8) , B (-3) : external outpost in ratio 3:4:****Choose the correct option.**

A 23

B -23 C -22 D -24

Classified Practice

Coordinate Geometry Basics

Q040 Challenge**Find pods A (-8) , B (-3) : midpoint supply hub:****Choose the correct option.**A $-13/2$ B $11/2$ C $-11/2$ D $-9/2$

Classified Practice

Coordinate Geometry Basics

Q041 Written

Point a has coordinates (2,3): determine its quadrant

Q042 Written**Point b has coordinates (2,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q043 **Written****Point c has coordinates (-2,3): determine its quadrant**

Q044 Written**Point d has coordinates $(-2, -3)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q045 **Written****Point a has coordinates (-4,2): determine its quadrant**

Q046 Written**Point b has coordinates (1,2): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant**

Q048 Written**Point d has coordinates $(-3,-1)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q049 **Written****Point a has coordinates (-3,1): determine its quadrant**

Q050 Written**Point b has coordinates (2,-8): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q051 Written

Point c has coordinates (4,4): determine its quadrant

Q052 Written**Point d has coordinates $(-6, -4)$: determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q053 Written

Point e has coordinates (-2,6): determine its quadrant

Q054 Written**Point f has coordinates (3,-3): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q055 Written

Point g has coordinates (-5,-7): determine its quadrant

Q056 Written**Point h has coordinates (1,4): determine its quadrant**

Classified Practice

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q058 MCQ**Point b has coordinates (2,-3): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates (-2,3): determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q060 MCQ**Point d has coordinates $(-2,-3)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates $(-4,2)$: determine its quadrant:**Choose the correct option.**A *II*B *I*C *III*D *IV*

Q062 MCQ**Point b has coordinates (1,2): determine its quadrant:****Choose the correct option.****A** *III***B** *IV***C** *I***D** *II*

Classified Practice

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Choose the correct option.**A *III*B *IV*C *I*D *II*

Q064 MCQ**Point d has coordinates $(-3,-1)$: determine its quadrant:****Choose the correct option.****A** *III***B** *I***C** *II***D** *IV*

Classified Practice

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates (-3,1): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q066 MCQ**Point b has coordinates (2,-8): determine its quadrant:****Choose the correct option.****A** *IV***B** *I***C** *II***D** *III*

Classified Practice

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Choose the correct option.**A *I*B *II*C *III*D *IV*

Q068 MCQ**Point d has coordinates $(-6, -4)$: determine its quadrant:****Choose the correct option.****A** *II***B** *IV***C** *III***D** *I*

Classified Practice

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates (-2,6): determine its quadrant:**Choose the correct option.**A *IV*B *II*C *I*D *III*

Q070 MCQ**Point f has coordinates (3,-3): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

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Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates (-5,-7): determine its quadrant:**Choose the correct option.**A *II*B *IV*C *III*D *I*

Q072 MCQ**Point h has coordinates (1,4): determine its quadrant:****Choose the correct option.****A** *II***B** *III***C** *IV***D** *I*

Classified Practice

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q075 **Written**

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Q076 Written

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q077 **Written**

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Q078 MCQ**For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.**A $(-3, 2)$ B $(3, -2)$ C $(-3, -2)$ D $(-2, -3)$

Classified Practice

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-2, -1)$

B $(-1, 2)$

C $(1, -2)$

D $(-1, -2)$

Q080 MCQ**For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-4, -2)$** **B $(4, 2)$** **C $(-4, 2)$** **D $(2, -4)$**

Classified Practice

Coordinate Geometry Basics

Q081 MCQ

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(2, 1)$

B $(1, -2)$

C $(-1, 2)$

D $(1, 2)$

Q082 MCQ**For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis:****Choose the correct option.****A $(-3, 4)$** **B $(3, -4)$** **C $(-3, -4)$** **D $(-4, -3)$**

Classified Practice

Coordinate Geometry Basics

Q083 Written

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q085 **Written**

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q086 Written

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q087 **Written**

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Q088 MCQ**For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Classified Practice

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(1, -2)$

B $(-1, 2)$

C $(1, 2)$

D $(2, 1)$

Q090 MCQ**For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(4, -2)$** **B $(-2, 4)$** **C $(4, 2)$** **D $(-4, -2)$**

Classified Practice

Coordinate Geometry Basics

Q091 MCQ

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-1, -2)$

B $(-2, -1)$

C $(-1, 2)$

D $(1, -2)$

Q092 MCQ**For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:****Choose the correct option.****A $(3, 4)$** **B $(4, 3)$** **C $(3, -4)$** **D $(-3, 4)$**

Classified Practice

Coordinate Geometry Basics

Q093 Written

For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.

Q094 Written

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q095 **Written**

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Q096 Written

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q097 **Written**

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.

Q098 MCQ**For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Choose the correct option.**A $(-3, -2)$ B $(3, -2)$ C $(-2, 3)$ D $(3, 2)$

Classified Practice

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, -2)$

B $(1, -2)$

C $(-2, 1)$

D $(1, 2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 2)$

B $(4, 2)$

C $(2, 4)$

D $(4, -2)$

Classified Practice

Coordinate Geometry Basics

Q101 MCQ

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-1, 2)$

B $(2, -1)$

C $(-1, -2)$

D $(1, 2)$

Q102 MCQ

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(-4, 3)$

B $(3, 4)$

C $(-3, -4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Classified Practice

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Classified Practice

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Classified Practice

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Choose the correct option.

A $(-3, -4)$

B $(-4, -3)$

C $(-3, 4)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Choose the correct option.

A $(-3, 4)$

B $(3, 4)$

C $(4, 3)$

D $(3, -4)$

Classified Practice

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Choose the correct option.

A $(3, 4)$

B $(-3, -4)$

C $(3, -4)$

D $(-4, 3)$

Classified Practice

Coordinate Geometry Basics

Q109 **Written**

Distance between A (2, 6) and B (5, 10)

Q110 Written**Distance between A (4, 3) and B (1, 2)**

Classified Practice

Coordinate Geometry Basics

Q111 Written

Distance between $A(\sqrt{3}, -1)$ and $B(0, 1)$

Q112 Written**Distance between A (1, 2) and B (11, 7)**

Classified Practice

Coordinate Geometry Basics

Q113 Written

Distance between A (3, -1) and B (-1, 3)

Q114 MCQ**Find distance between A (2, 6) and B (5, 10):****Choose the correct option.****A** 4**B** -5**C** 5**D** 6

Classified Practice

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Choose the correct option.**

A $\sqrt{10}$

B $2\sqrt{10}$

C (10)

D 1

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Choose the correct option.**

A (7)

B 1

C $\sqrt{7}$ D $2\sqrt{7}$

Classified Practice

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Choose the correct option.**

A 1

B $5\sqrt{5}$ C $52\sqrt{5}$ D $5(5)$

Q118 MCQ**Find distance between A (3, -1) and B (-1, 3):****Choose the correct option.**

A $4\sqrt{2}$

B $42\sqrt{2}$

C $4(2)$

D 1

Classified Practice

Coordinate Geometry Basics

Q119 **Written****A (3, -2), B (x, 1), distance 5: find x**

Q120 Written

A $(-1, 3)$, **B** $(x, 1)$, distance $\sqrt{5}$: find **x**

Classified Practice

Coordinate Geometry Basics

Q121 **Written****A (3, -1), B (x, 2), distance 5: find x**

Q122 Written

A (1, -3), **B** (-2, y), **distance** $\sqrt{13}$: **find** y

Classified Practice

Coordinate Geometry Basics

Q123 Written

A $(-4, -2)$, B $(x, 3)$, distance $5\sqrt{5}$: find x

Q124 MCQ

A $(3, -2)$, **B** $(x, 1)$, **distance 5: find x:**

Choose the correct option.

A $x = -1, 7$ (one value only)

B $x = -1, 7$

C $x = +1, 7$

D $x = -17$

Classified Practice

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, B $(x, 1)$, distance $\sqrt{5}$: find x:

Choose the correct option.

A $x = -20$

B $x = -2, 0$ (one value only)

C $x = -2, 0$

D $x = +2, 0$

Q126 MCQ

A $(3, -1)$, **B** $(x, 2)$, **distance 5: find x:**

Choose the correct option.

A $x = -1, 7$ (one value only)

B $x = -1, 7$

C $x = +1, 7$

D $x = -17$

Classified Practice

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y:

Choose the correct option.

A $y = -5, -1$

B $y = +5, -1$

C $y = -5 - 1$

D $y = -5, -1$ (one value only)

Q128 MCQ**A** $(-4, -2)$, **B** $(x, 3)$, **distance** $5\sqrt{5}$: **find x:****Choose the correct option.****A** $x = +14, 6$ **B** $x = -146$ **C** $x = -14, 6$ *(one value only)***D** $x = -14, 6$

Classified Practice

Coordinate Geometry Basics

Q129 Written

Distance from origin to A (2, -4)

Q130 Written**Distance from origin to A $(-3, 4)$**

Classified Practice

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Choose the correct option.**A $2(5)$

B 1

C $2\sqrt{5}$ D $22\sqrt{5}$

Q132 MCQ

Find distance from origin to A $(-3, 4)$:**Choose the correct option.**

A 4

B -5

C 5

D 6

Classified Practice

Coordinate Geometry Basics

Q133 **Challenge****Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x**

Classified Practice

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x :

Choose the correct option.

A $x = -23$

B $x = -2, 3$ (one value only)

C $x = -2, 3$

D $x = +2, 3$

Classified Practice

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A (−2, 1) and B (3, 4)**

Q136 Written**Point on the y-axis equidistant from A (4, 2) and B (1, -1)**

Classified Practice

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$**

Q138 Written**Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$**

Classified Practice

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$**

Q140 MCQ**Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Choose the correct option.**

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Q142 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$:****Choose the correct option.**A $(-2, 0)$ B $(-3, 0)$ C $(2, 0)$ D $(0, 2)$

Classified Practice

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Choose the correct option.**A $(0, -5/6)$ B $(-1, 5/6)$ C $(0, 5/6)$ D $(5/6, 0)$

Q144 MCQ**Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:****Choose the correct option.**A $(-19/2, 0)$ B $(17/2, 0)$ C $(0, 17/2)$ D $(-17/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)**

Q146 Written**Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)**

Classified Practice

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)**

Q148 Written**Point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$**

Classified Practice

Coordinate Geometry Basics

Q149 **Written****Point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6)**

Q150 MCQ**Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Choose the correct option.****A** $(-13/8, 9/4)$ **B** $(13/8, 9/4)$ **C** $(9/4, 13/8)$ **D** $(13/8, -9/4)$

Classified Practice

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Choose the correct option.**

A (-4, 11)

B (4, -11)

C (-4, -11)

D (-11, -4)

Q152 MCQ**Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):****Choose the correct option.****A** (2, 1)**B** (1, 2)**C** (2, -1)**D** (-2, 1)

Classified Practice

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Choose the correct option.**A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

Q154 MCQ

Find point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6):**Choose the correct option.**

A (2, 3)

B (−2, 3)

C (3, −2)

D (−2, −3)

Classified Practice

Coordinate Geometry Basics

Q155 **Challenge**

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Classified Practice

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with $CA=CB$ and D (x, 0) with $DA=DB$: find CD:

Choose the correct option.

A $3\sqrt{5}$

B $32\sqrt{5}$

C 3(5)

D 1

Classified Practice

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Classified Practice

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Classified Practice

Coordinate Geometry Basics

Q160 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q161 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Classified Practice

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(4/3, -11/3)$

B $(-4/3, 11/3)$

C $(4/3, 11/3)$

D $(11/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Choose the correct option.

A $(-15/4, -13/4)$

B $(15/4, 13/4)$

C $(-15/4, 13/4)$

D $(13/4, -15/4)$

Classified Practice

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Choose the correct option.

A (2, 9/5)

B (-2, -9/5)

C (2, -9/5)

D (-9/5, 2)

Classified Practice

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Choose the correct option.

A $(-3, 3)$

B $(-3, -3)$

C $(3, 3)$

D $(3, -3)$

Classified Practice

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Choose the correct option.

A $(-2/3, -5/3)$

B $(-5/3, -2/3)$

C $(-2/3, 5/3)$

D $(2/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q167 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q169 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Classified Practice

Coordinate Geometry Basics

Q170 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Classified Practice

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Classified Practice

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(-2, 19/2)$

B $(2, 19/2)$

C $(19/2, 2)$

D $(2, -19/2)$

Classified Practice

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A $(11, 18)$

B $(-11, -18)$

C $(11, -18)$

D $(-18, 11)$

Classified Practice

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Choose the correct option.

A (-10, -15)

B (10, -15)

C (-15, 10)

D (10, 15)

Classified Practice

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Choose the correct option.

A $(8, -11)$

B $(-8, -11)$

C $(-11, -8)$

D $(-8, 11)$

Classified Practice

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Choose the correct option.

A $(0, 15/2)$

B $(-1, -15/2)$

C $(0, -15/2)$

D $(-15/2, 0)$

Classified Practice

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Classified Practice

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Classified Practice

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q185 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Classified Practice

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Classified Practice

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Choose the correct option.

A $(3/2, -1/2)$

B $(-1/2, 3/2)$

C $(3/2, 1/2)$

D $(-3/2, -1/2)$

Classified Practice

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 2)$

B $(-2, -2)$

C $(2, 2)$

D $(2, -2)$

Classified Practice

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(0, 0)$

B $(-1, 0)$

C $(0, -1)$

D $(1, 0)$

Classified Practice

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(-2, 1)$

B $(1, -2)$

C $(-2, -1)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Choose the correct option.

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

Classified Practice

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A (1, 2/3)

B (2/3, -1)

C (-2/3, 1)

D (2/3, 1)

Classified Practice

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Choose the correct option.

A $(-2, -1)$

B $(2, -1)$

C $(-1, 2)$

D $(2, 1)$

Classified Practice

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A $(5/3, 5/3)$

B $(5/3, -5/3)$

C $(-5/3, 5/3)$

D $(-5/3, -5/3)$

Classified Practice

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Choose the correct option.

A $(-2, -5/2)$

B $(2, -5/2)$

C $(-5/2, 2)$

D $(2, 5/2)$

Classified Practice

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Choose the correct option.

A 1

B 2

C $(0, 0)$ D $(-1, 0)$

Classified Practice

Coordinate Geometry Basics

Q197 **Challenge****A $(-2, 3)$, B $(6, 1)$, C $(0, y)$, $CA=CB$: find the centroid G**

Classified Practice

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB**: find the centroid **G**:

Choose the correct option.

A $(4/3, 2/3)$

B $(-4/3, -2/3)$

C $(4/3, -2/3)$

D $(-2/3, 4/3)$

Classified Practice

Coordinate Geometry Basics

Q199 **Written****Point with A $(-3, 2)$ symmetric about centre $(2, 6)$**

Q200 Written**Point with A (1, -2) symmetric about centre (2, 3)**

Classified Practice

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)**

Q202 Written**Point with A $(-3, 1)$ symmetric about centre $(2, 0)$**

Classified Practice

Coordinate Geometry Basics

Q203 **Written**

Point with A (7, 4) symmetric about centre (-3, 5)

Q204 Written**Point with A (4, -8) symmetric about centre (-3, 2)**

Classified Practice

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)**

Q206 Written**Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$**

Classified Practice

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Choose the correct option.**A $(7, -10)$ B $(-7, 10)$ C $(7, 10)$ D $(10, 7)$

Q208 MCQ**Find point with A (1, -2) symmetric about centre (2, 3):****Choose the correct option.**

A (-3, 8)

B (3, 8)

C (8, 3)

D (3, -8)

Classified Practice

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Choose the correct option.**

A (-5, 8)

B (8, -5)

C (-5, -8)

D (5, 8)

Q210 MCQ**Find point with A $(-3, 1)$ symmetric about centre $(2, 0)$:****Choose the correct option.**A $(7, 1)$ B $(-7, -1)$ C $(7, -1)$ D $(-1, 7)$

Classified Practice

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Choose the correct option.**

A (-13, -6)

B (13, 6)

C (-13, 6)

D (6, -13)

Q212 MCQ**Find point with A (4, -8) symmetric about centre (-3, 2):****Choose the correct option.**

A (-10, -12)

B (10, 12)

C (-10, 12)

D (12, -10)

Classified Practice

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Choose the correct option.**

A (8, 6)

B (-8, 6)

C (6, -8)

D (-8, -6)

Q214 MCQ**Find point with A $(-4, 2)$ symmetric about centre $(-1, -3)$:****Choose the correct option.**A $(2, -8)$ B $(-8, 2)$ C $(2, 8)$ D $(-2, -8)$

Classified Practice

Coordinate Geometry Basics

Q215 **Challenge****B (m , 5), C (4, n) symmetric about A (1, -1): find $m+n$**

Classified Practice

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Choose the correct option.

A -10

B 9

C -9

D -8

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Point A $(-3, 4)$ lies in which quadrant?:

Choose the correct option.

A *II*

B *I*

C *III*

D *IV*

Quiz MCQ 2 Concept Quiz · MCQ

Find the distance between A (1, 2) and B (4, 6):

Choose the correct option.

A -5

B 5

C 6

D 4

Classified Practice

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz](#) · MCQ

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Choose the correct option.

A $(0, -5/2)$

B $(-1, 5/2)$

C $(0, 5/2)$

D $(5/2, 0)$

Quiz MCQ 4 Concept Quiz · MCQ

Point B is symmetric to A (2, -1) about P (-3, 4). Find B:

Choose the correct option.

A (-8, -9)

B (8, 9)

C (-8, 9)

D (9, -8)

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Quiz WRITTEN 6 Concept Quiz · Written

For points A (-5) and B (7) , find the point dividing AB externally in ratio 2:5.

Classified Practice

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A $(-2, 1)$, **B** $(6, -3)$, **C** $(1, 7)$.

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, 5/6)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)